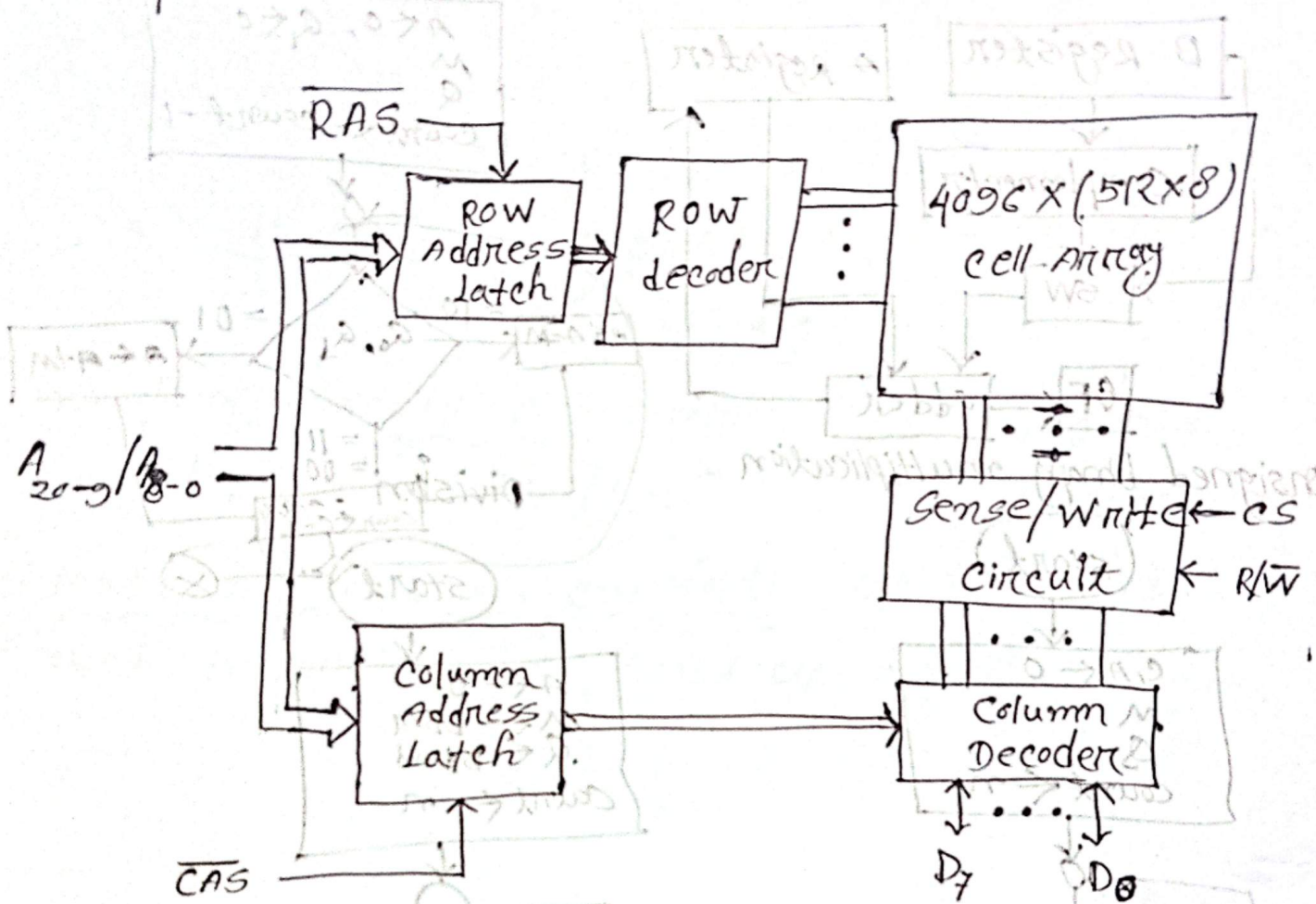
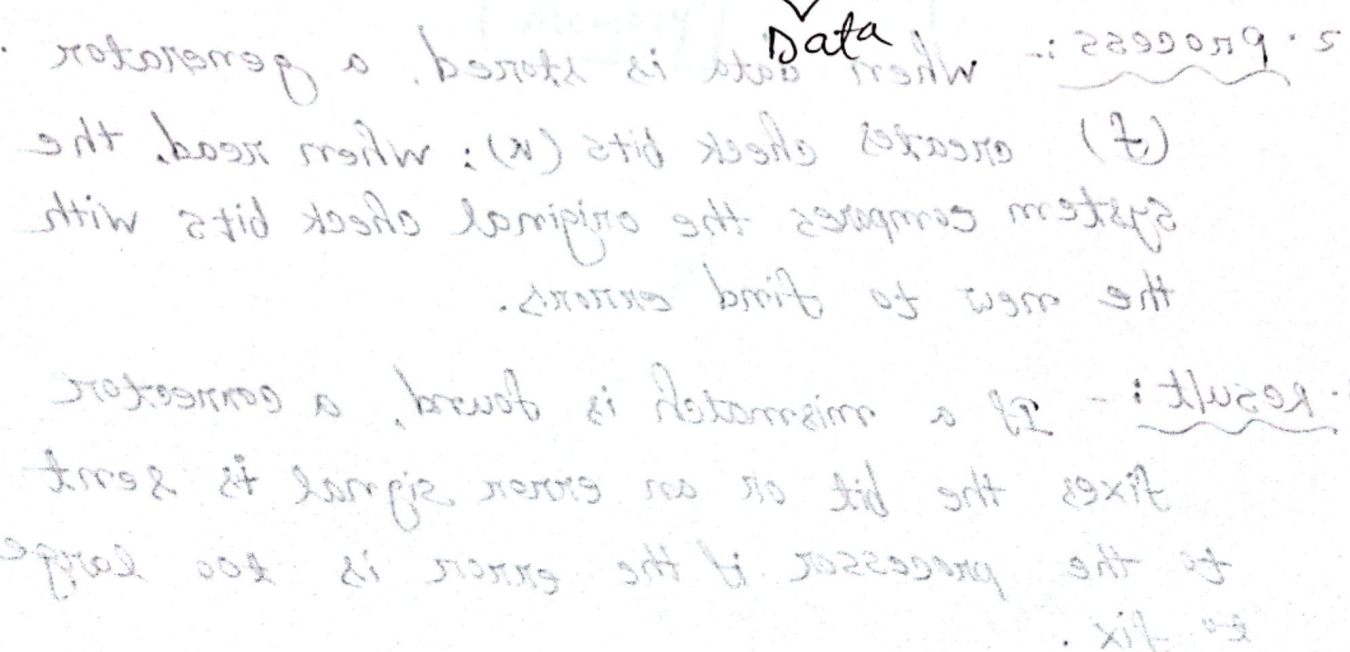


16 Megabit DRAM chip / $2M \times 8$ DRAM / Dynamic Memory chip.

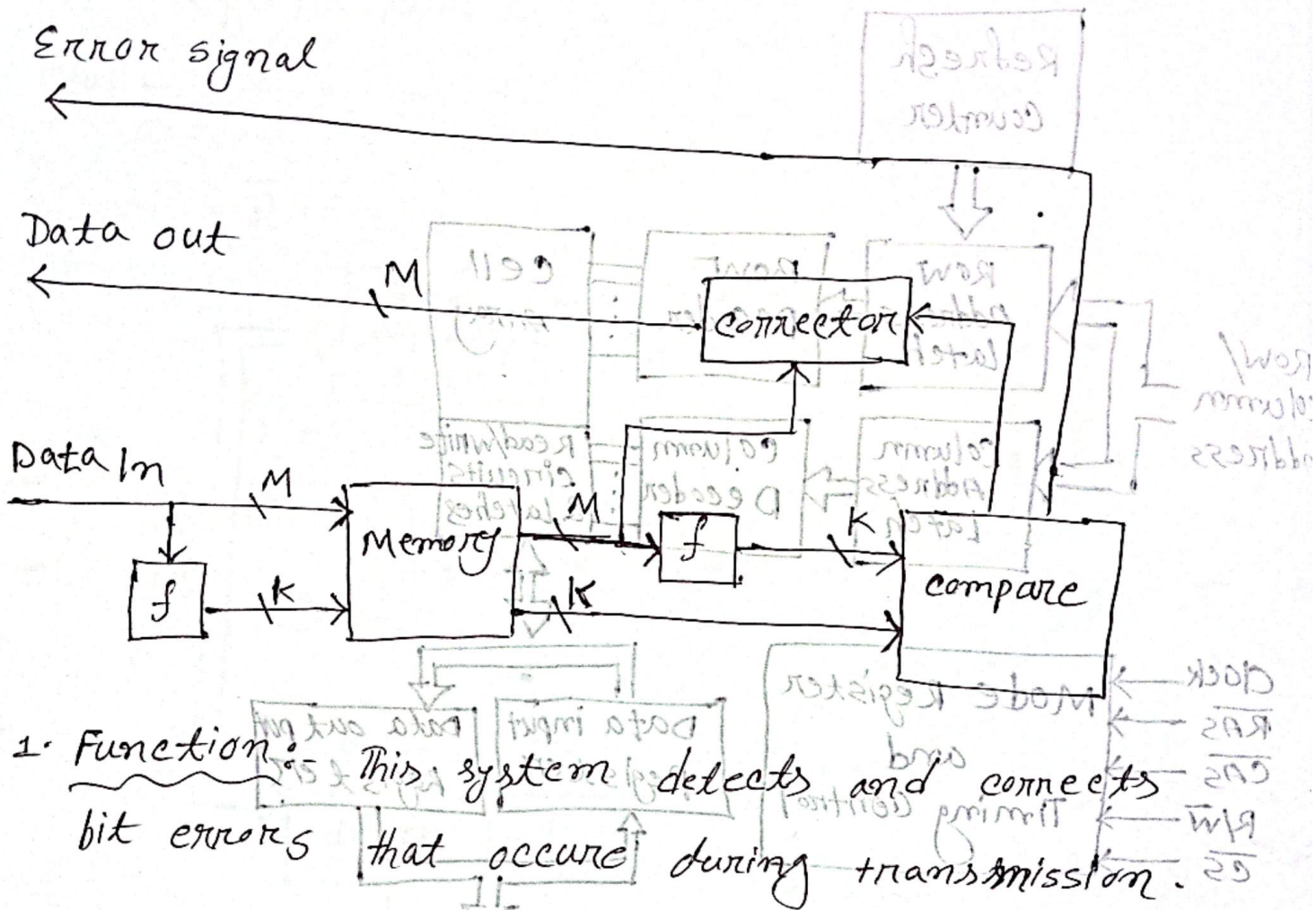


1. Organization: It stores 16 Megabits of data organized as $2M$ (rows/columns) $\times$ 8 bits (word size).
2. Address Multiplexing: To save pins, it uses Row address strobe (**RAS**) and Column address strobe (**CAS**) to latch address part separately.
3. Components: It consists of cell array, a row decoder to select the line and a sense/write circuit to read or store the 8 bit data.



Error correcting Diagram:-

RAM :-

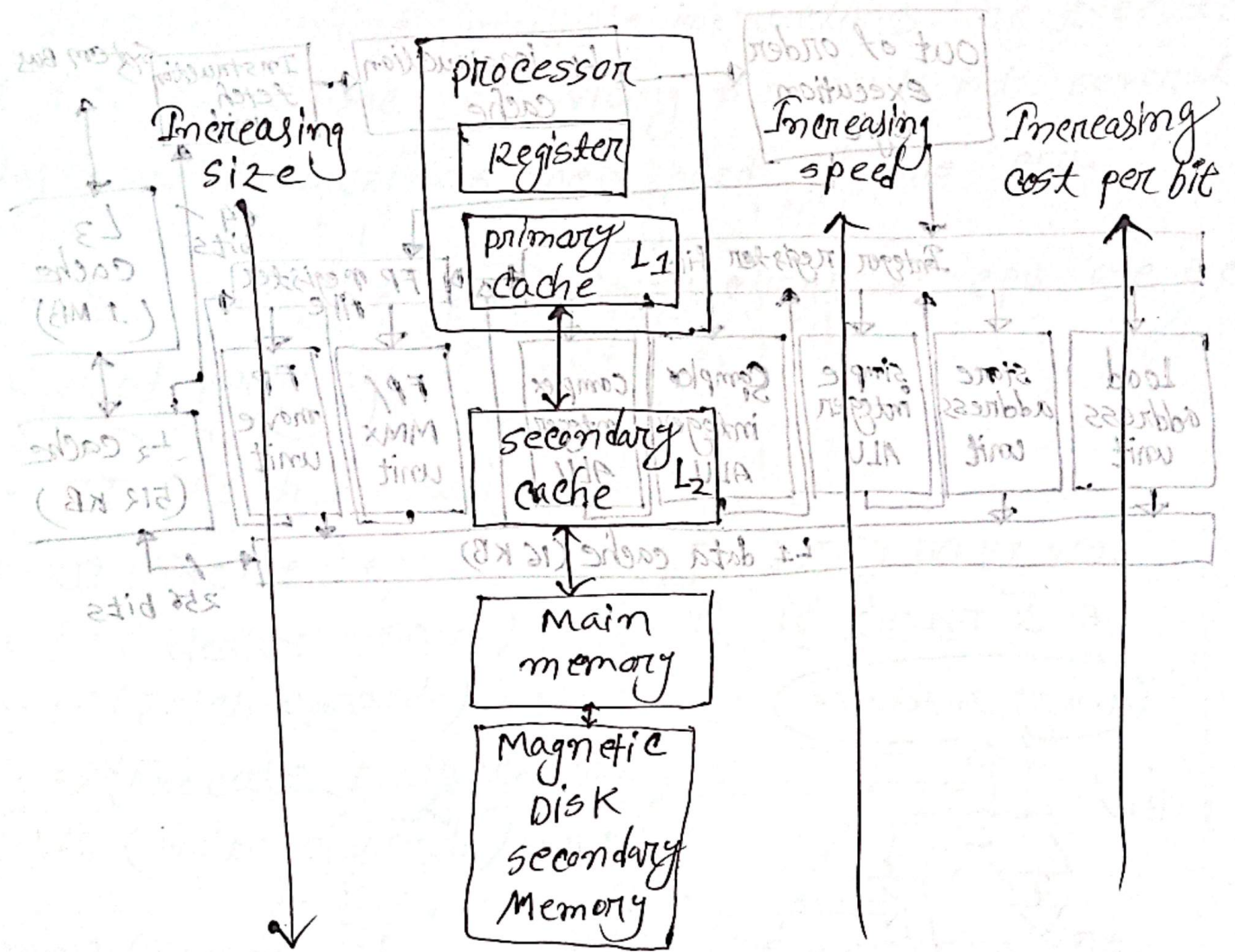


1. Function: This system detects and connects bit errors that occur during transmission.

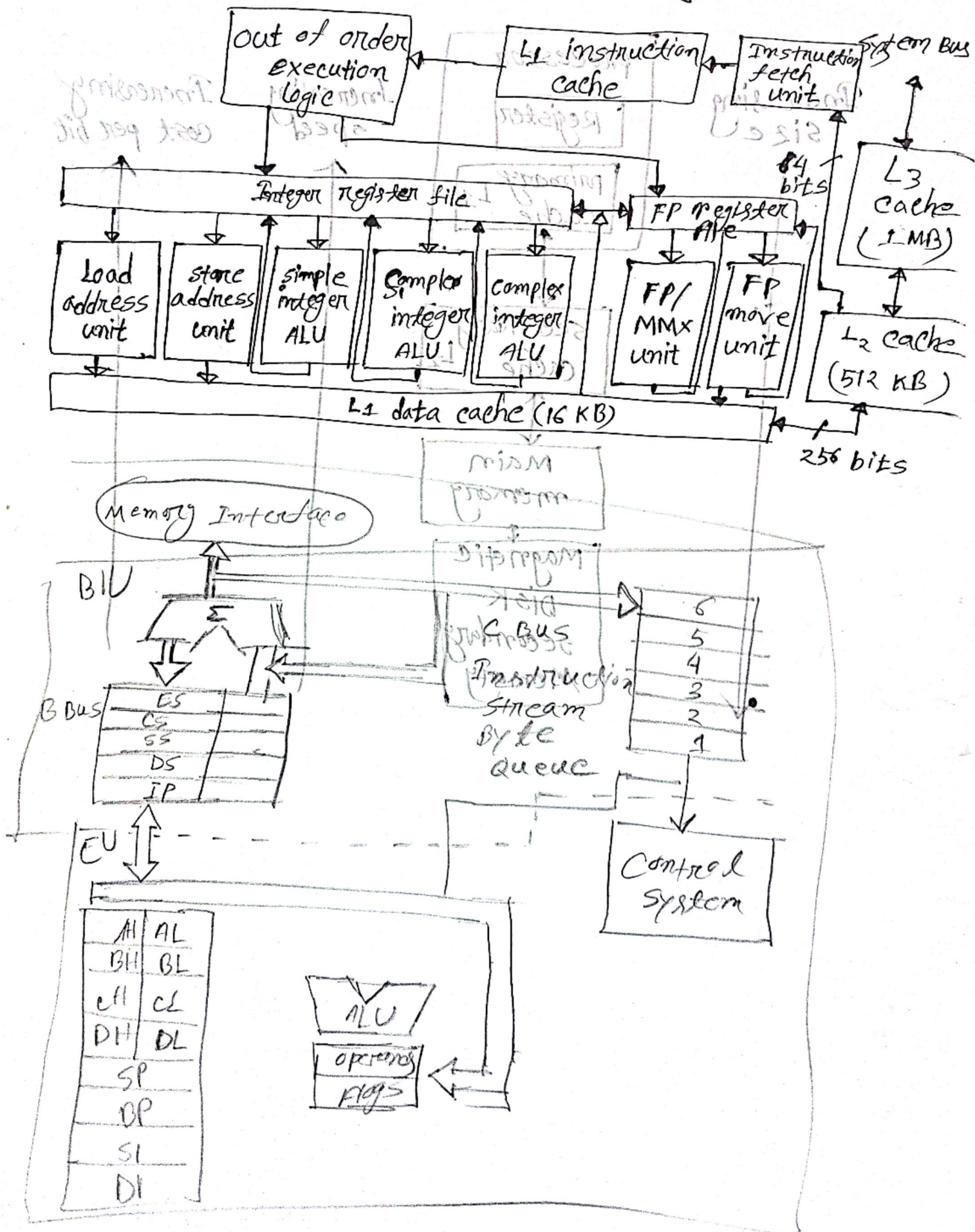
2. process: - when data is stored, a generator (f) creates check bits (k); when read, the system compares the original check bits with the new to find errors.

3. Result: - If a mismatch is found, a connector fixes the bit or an error signal is sent to the processor if the error is too large to fix.

Memory hierarchy (size, speed, cost)



*Diagram of pentium 4 block Diagram:-



Pipelining:- Pipelining is a technique in computer architecture where multiple instructions are executed at the same time by dividing the work into several stages, which increases the speed of the CPU.

Instruction pipelining:- multiple instructions of CPU are executed

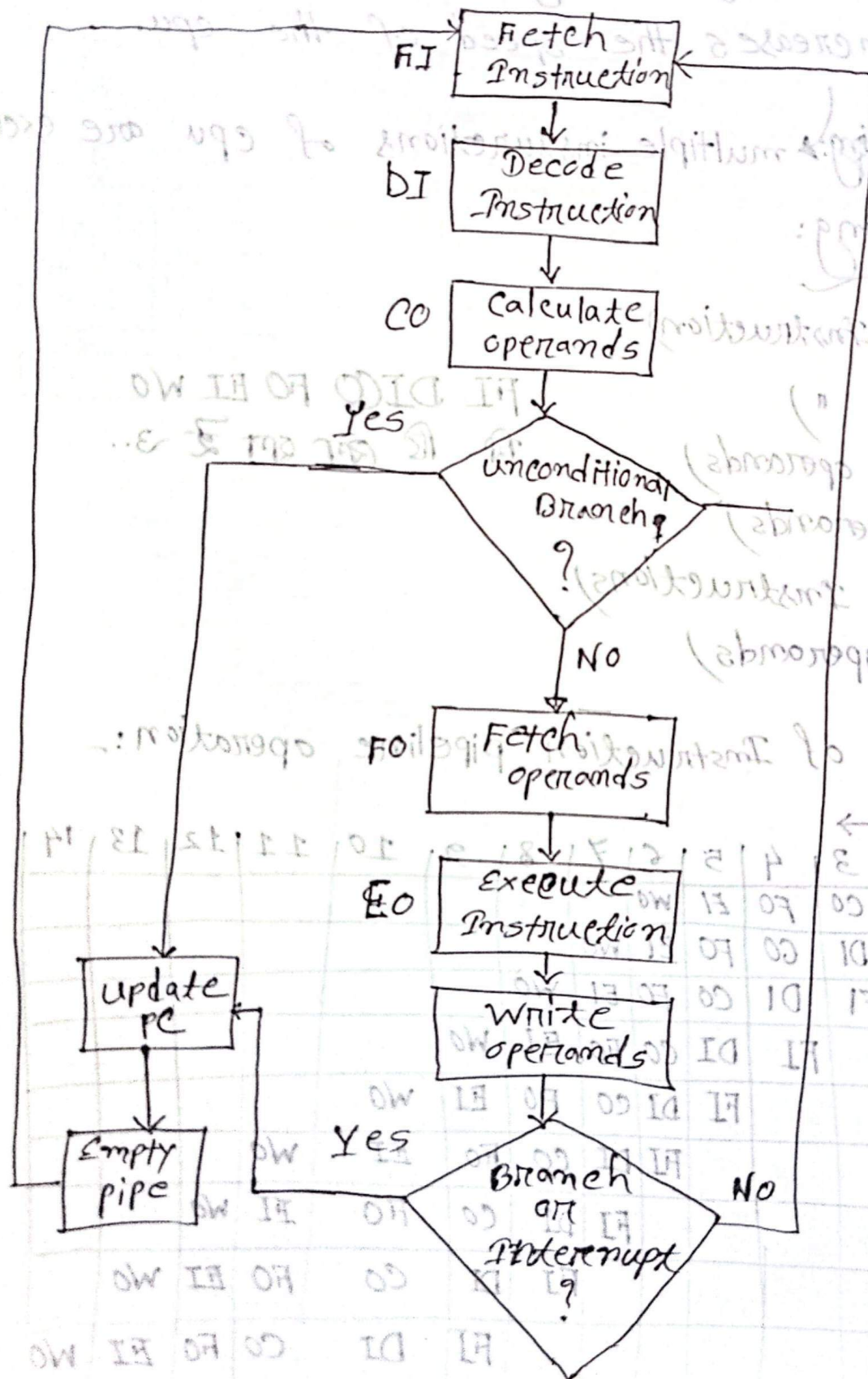
stages of pipelining:

1. FI (Fetch Instruction)
2. DI (Decode ")
3. CO (Calculate operands)
4. FO (Fetch operands)
5. EX (Execute Instructions)
6. WO (write operands)

Timing Diagram of Instruction pipeline operation:-

	Time →													
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	FI	DI	CO	FO	EI	WO								
2		FI	DI	CO	FO	EI	WO							
3			FI	DI	CO	FO	EI	WO						
4				FI	DI	CO	FO	EI	WO					
5					FI	DI	CO	FO	EI	WO				
6						FI	DI	CO	FO	EI	WO			
7							FI	DI	CO	FO	EI	WO		
8								FI	DI	CO	FO	EI	WO	
9									FI	DI	CO	FO	EI	WO

▣ Six (6) stage pipeline Diagram/ Instruction



☐ Hazard:-

A hazard is a problem that happens in a "pipelined" processor. It occurs when the next instruction cannot start or finish in the very next step in the following clock cycles that causes a possible or potential delay and breaks the smooth flow of work as a result the computer runs slower.

☐ Stall:-

A stall is simply a delay of time during the execution of an instruction. In a pipeline, this is often called a bubble.

Types of Hazard:-

structural:- A structural Hazard occurs when there are not enough hardware parts to handle all the instructions trying to run at the same time. For example, a conflict arises if two different instructions both try to use the memory during same clock cycle.

Data:- A data Hazard happens when instructions that depend on each other but try to change data at different stages in the pipeline.

Control:- A control hazard comes from "branch" instructions that change the path of the program these instructions change the program counter, that

tells the program counter computer what to do next.

Solutions of Data Hazard:-

To solve data hazard, the processor uses forwarding, that sends the result directly to the next instruction instead of waiting to store it in memory. If the data is ready not ready, the hardware uses an Interlock to pause the pipeline until the needed data becomes available is ready.

Another method is Instruction scheduling where instructions are rearranged so that dependent instructions do not come together, helping the pipeline run smoothly.

Data:- A data hazard happens when instructions that depend on each other but try to change data at different steps in the pipeline.

Control:- A control hazard comes from "branch" instructions that change the path of the program. These instructions change the program counter, that

Solutions of Structural Hazard:-

To solve the structural hazard, the processor uses "Resource Duplication", that means adding extra hardware so that instruction cannot compete for the same resource.

Another method is "splitting memory" into two parts: one for instructions and one for data. This helps the processor to fetch instructions and data at the same time without delay.

Solution of Control Hazard:

To solve control hazard, the processor uses Branch prediction to guess the result of a branch and continue processing. A "Branch Target Buffer (BTB)" stores previous branch addresses to help prediction.

"Delayed Branching": runs other useful instructions during the waiting time. Compiler reorders instruction so that useful work is done in delay slot.

"Speculative Execution" follows the predicted path and keeps the result only if the prediction is correct.

Describe 3 Replacement Algorithms with example

① FIFO (First In First out):

In this algorithm, the ~~page~~ ^{block} entered that entered memory first is removed first as a new ~~page~~ ^{block} need space.

Example:

If ~~pages~~ ^{blocks} in memory are 1, 2, 3 and a new page 4 arrives, ~~page~~ ^{block} 1 will be replaced as it came first.

② LRU (Least Recently Used):

This algorithm removes the ~~page~~ ^{block} that has not been used for the longest time.

Example:

If ~~pages~~ ^{blocks} 1, 2, 3 are in memory and ~~page~~ ^{block} 1 and 3 were used recently but 2 was not used for a long time, then 2 will be replaced by the new ~~page~~ ^{block}.

③ LFU (Least Frequently used):

This algorithm replaces the ~~blocks~~ ^{block} that has been used the least number of time.

Example:

If block 1 is used 5 times, block 2 2 times block 3 1 time, then 3 will be replaced.

printer

* printer :- A printer is an electromechanical devices that converts the text and graphical documents from electronic form to the physical form.

Impact printer

- i. Daisy wheel
- ii. Dot-matrix

No-Impact printer

- i. Inkjet
- ii. Thermal
- iii. Laser

* Impact printer vs Non-Impact printer :-

Feature	Impact printer	Non-Impact Printer
Printing Method	Prints by striking the paper using pins or hammers	prints without striking the paper.
Noise	Generate high level of noise, while printing	Doesn't generate any noise, while printing
Speed	printing speed is slow and consume lot of time	printing speed very fast, can print lot of page per minute.
Quality	printing quality is lower compared to non-impact.	quality is higher than impact
Cost	They are expensive	They are more expensive
paper sheet used	They use continuous paper sheet.	They use individual paper sheet often.
Example	Daisy wheel, Dot-Matrix	Laser, Thermal, Inkjet

Daisy wheel printer:-

Adv:-

- i. It is cheap to buy and maintain
- ii. Creates carbon copies.
- iii. Low maintenance
- iv. Low cost

Dis-Adv:-

- i. speed is very slow
- ii. Can't print graphics
- iii. limited number of characters
- iv. noisy.

Dot-Matrix printer:-

Adv:-

- i. print multiple carbon copies
- ii. Low printing cost per page
- iii. used for continuous form papers
- iv. Reliable - durable

Dis-Adv:-

- i. Noisy
- ii. Limited print quality
- iii. low speed
- iv. Limited color printing.

Inkjet printer:-

Ad:-

- (i) low cost
- (ii) High quality of output
- (iii) capable of printing picture
- (iv) Easy to use and printing speed is fast

dis-Ad:-

- (i) Less durable, and chance to easily damage
- (ii) Maintenance cost is expensive
- (iii) Not good for high volume printing

Thermal printer:-

Ad:-

- (i) No ink is needed
- (ii) Fast printing speed
- (iii) Smooth and silent
- (iv) Low maintenance
- (v) Small size easy to carry

dis-Ad:-

- (i) needs Thermal paper
- (ii) Not suitable for long time use
- (iii) Limited color printing
- (iv) High temperature may damage the print.

Laser printer :-

Adv:-

- (i) More cost effective than inkjet and thermal printer.
- (ii) Increase productivity
- (iii) High printing speed
- (iv) Higher paper capacity
- (v) support multi-color

dis-Adv:-

- (i) Initial investment cost is high
- (ii) Ink is more expensive than inkjet.
- (iii) Require short 'warm-up' times
- (iv) Costly as support multi-color.

Keyboard

Types of keyboard :

→ Mechanical key switches

→ Membrane key switches

→ Capacitive key switches

→ Hall effect key switches

Mechanical key switches

1. Two pieces of metal pushed together when a key is pressed.

2. Contact bounce.

3. Get oxidized or dirty.

4. Lifetime :

1 million key stroke

→ Lifetime 20 million key strokes

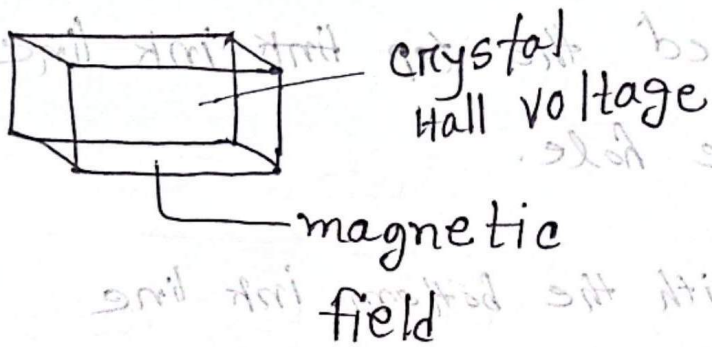
Hall effect keyswitch:

→ A reference current is passed through a semiconductor crystal, between two opposing faces.

→ When a key is pressed, the crystal is moved through a magnetic field which has flux lines perpendicular to the current flow.

→ It produces a small voltage called Hall voltage.

→ Very dependable and 100 million key strokes



Capacitive key switches:-

Structure: It consists of two metal plates. one is movable, another is fixed.

Action: When a key is pressed the movable plate is pushed closer to the fixed plate.

Detection: This change is detected by a circuit that produces a logic signal.

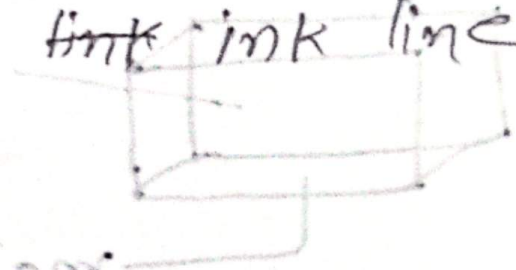
Advantage: It is highly durable as no physical contact.

Life time: 20 million keystrokes.

Membrane key switches:

- It is a special type of mechanical switch.
- It consists of three layers of rubber or plastic like a sandwich.
- Top layer has a conducting line of silver ink that runs under each ~~layer~~ row of keys.
- Middle layer has a hole under each key act as a spacer.
- Bottom layer has a conducting line of silver ink runs under column of key.

• when a key is pressed the top ink line is pushed through the hole.



• It makes contact with the bottom ink line to complete the circuit.

∴ positive red switch